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Unravelling Inter-Organisational Collaboration to Address the Circular Economy Transitions: A Systematic Literature Review

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ABSTRACT

This study offers a systematic review of the growing body of literature about inter-organisational collaboration initiatives that have arisen to deal with the circular economy transition. Indeed, circular economy transition is highly reliant on stakeholders' interaction in collaborative patterns, where organisations bring together specific knowledge and resources to overcome technical and economic constraints. Building on this awareness, an increasing number of studies furthered the understanding of circular economy from a meso-level perspective and the need for systematisation has become evident. In response to the lack of general conceptualisation and characterisation of inter-organisational collaboration initiatives, this article provides a clearer picture of the most common collaborative approaches discussed in the circular economy literature. As a result of the bibliometric and thematic content analysis of 78 scientific papers, we illustrate three thematic clusters and outline five dimensions to put forward a novel conceptual compass for circular economy ventures. Delving into circular partnerships, supply chains, and ecosystems, we offer a first systematisation of such a flourishing literature stream and suggest future avenues by handling several research questions for forthcoming studies. In terms of managerial implications, this study supports decision-making by comparing different strategic approaches to address the circular economy transition in collaboration with firm's stakeholders.

> *Humanity has a choice: cooperate or perish.*
> Antonio Guterres, United Nations Secretary-General
> 2022 United Nations Annual Climate Conference (COP27)

1 | Introduction

Pressing environmental challenges and global resource depletion raise the need for sustainable solutions that can underpin a balanced economic model. As stressed by leading policy bodies (e.g., European Commission 2020; United Nations 2024; United Nations Environment Programme 2024), take-make-use-dispose production systems undermine the co-occurrence of economic growth with environmental preservation and human well-being. In response, circular economy (CE) gains ground as a holistic, transformative, and restorative approach to achieve sustainable development (Hina et al. 2023; Murray et al. 2017; Triguero et al. 2022). Based on the awareness of planetary resource limits, as well as the vision of a world as a system in which different actors collaborate to reduce pollution and waste, CE is intended to maximise the natural ecosystem functioning in a 'cradle-to-cradle' economy (Ellen MacArthur Foundation 2013; Kanda et al. 2021; McDonough and Braungart 2010). This innovative economic model is firmly endorsed by leading policy bodies such as the United Nations (2024) and the European

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Commission (2020), besides being fuelled by end consumers' growing sensitivity towards sustainable consumption (see McKinsey and Company 2023; PricewaterhouseCoopers 2024; Reichheld et al. 2023). As a result, more and more companies are committing to CE principles implementation by translating environmental and social purposes into business practices (Hopkinson et al. 2018; Perotti et al. 2024; Ranta et al. 2018; Sohal and De Vass 2022).

This restorative and holistic approach to corporate sustainability requires business models' innovation to integrate CE principles in the dynamics of value proposition, value capture, value creation and delivery (Ferasso et al. 2020; Lüdeke-Freund et al. 2019; Pieroni et al. 2019; Santa-Maria et al. 2022). As outlined by Bocken et al. (2016) and Geissdoerfer et al. (2018), CE integration in businesses is commonly associated with innovative strategies undertaken to (a) reduce the overall volume of resources and energy required to produce goods and services (i.e., narrowing); (b) extend the life cycle of products by decreasing the frequency of resource consumption (i.e., slowing); (c) create a closed-loop system where the outputs of one process become the inputs of another, effectively mimicking natural cycles (i.e., closing); (d) maximise the utility or productivity of resources within a given system (i.e., intensifying); or (e) decrease material content associated with products and services (i.e., dematerialising). Economic organisations are thus called to rethink their businesses, eventually by drawing upon systemic configurations that connect multiple actors (Hopkinson et al. 2018; Kanda et al. 2021; Köhler et al. 2022; Perotti et al. 2024; Zucchella and Previtali 2019). In this vein, previous studies observed that addressing the CE transition entails more than just single-business focused transformations (Ciulli et al. 2020; Konietzko et al. 2020; Moggi and Dameri 2021; Zaoual and Lecocq 2018). The innovation effort required to implement CE strategies in traditional (or new) businesses usually calls for knowledge and resources that go far beyond those possessed by isolated companies (Calabrese et al. 2024; Köhler et al. 2022; Perotti et al. 2025).

In line with a meso-level comprehension of CE (Barreiro-Gen and Lozano 2020; Bressanelli et al. 2022; Ghisellini et al. 2016; Murray et al. 2017), the importance of inter-organisational collaborations in succeeding CE transition was underscored by looking at stakeholder engagement and multidimensional value creation (Civera et al. 2025; Tapaninaho and Heikkinen 2022). Extant CE literature acknowledges stakeholder relationships' weight, for instance, by observing partnerships and cooperative networks that allow firms and other entities to share resources and knowledge, exchange by-products, or collectively recover waste by closing material loops (Moggi and Dameri 2021; Perotti et al. 2024). Accordingly, previous studies explored companies' aggregation in the form of coordination along the supply chain (Aarikka-Stenroos et al. 2022; Bressanelli et al. 2019; De Angelis et al. 2018; Tseng et al. 2022) and industrial districts (Bressanelli et al. 2022), industrial symbiosis (Baldassarre et al. 2019; Palagonia et al. 2025) and ecosystems (Nogueira et al. 2023; Kanda et al. 2021; Parida et al. 2019; Zucchella and Previtali 2019). Strategic alliances have also been recognised by Suchek et al. (2021) as a thematic area that deserves further exploration in extant CE and innovation literature. Köhler et al. (2022), Jesus and Jugend (2023), and Calabrese et al. (2024) emphasised the close link between CE and open innovation. Furthermore, scholars recommend dwelling on the interactions and synergies among heterogeneous stakeholders to ensure successful circular innovations (Brown et al. 2019; De Angelis et al. 2023; Perotti et al. 2025). Past research, such as Dangelico and Pontrandolfo (2015) and Yu et al. (2022), even emphasises how the collaborative strategies of circular businesses can represent the nexus between environmental management and firm performance. In sum, inter-organisational collaboration facilitates the integration of CE principles in businesses by leveraging external critical resources, knowledge, and capabilities to overcome barriers towards sustainable development (Aarikka-Stenroos et al. 2022; Galvão et al. 2022; Seles et al. 2022; Wade et al. 2022).

Accordingly, we infer that inter-organisational collaborations and stakeholder interactions are pivotal in achieving the CE transition (e.g., Aarikka-Stenroos et al. 2022; Jesus and Jugend 2023; Tapaninaho and Heikkinen 2022; Zucchella and Previtali 2019). This burgeoning literary niche demands a specific systematisation, however. Several distinct connotations of collaborative patterns in the CE literature converge to outline a murky understanding, which may lead to ambiguities in academics and practitioners' interpretation. A comprehensive assessment of previous systematic literature reviews (see Table 1) reveals an open research gap. Even though previous literature reviews attempted to explore the meso dimension of CE related to innovation management and strategic alliances (e.g., Sehnem et al. 2022; Suchek et al. 2021; Jesus and Jugend 2023) and circular supply chain management (e.g., Eisenreich et al. 2022; Lahane et al. 2020; De Angelis et al. 2018; Sudusinghe and Seuring 2022), extant studies fall short in terms of delivering an overarching systematisation of inter-organisational collaboration patterns that effectively support the CE transition. More specifically, a clarification of the stakeholders' nature and interaction mechanisms in collaborative formations is missing (e.g., Cooper and Claxton 2022; Pedersen et al. 2023; Perotti et al. 2024; Tapaninaho and Heikkinen 2022). Therefore, this study aims to review and systematise previous scientific contributions to gain insights into the organisational structure, relational ties, and scope of inter-organisational collaborations to deliver insight and guide future studies (e.g., Ferasso et al. 2020; Hassan et al. 2023; Khan, Kaur, et al. 2021; Madanaguli et al. 2021). The following research questions (RQs) have been raised accordingly.

RQ1: How has the business management literature addressed the issue of inter-organisational collaboration to achieve circular economy transition?

RQ2: What are the thematic areas of focal investigation in this regard?

RQ3: What are the main gaps in the existing body of literature and potential future avenues for advancing this research domain?

We employed a bibliometric analysis to answer RQ1, offering a detailed research profile and identifying the main literary streams pertaining to this literature domain. With regard to RQ2, we conducted a comprehensive content analysis to frame the collaborative patterns recognised from previous studies.

TABLE 1 | Most relevant systematic literature reviews in the circular economy domain.

Article information	Research questions/Objectives	Outcomes and implications	Level of analysis	Review method	Range of years
Jesus and Jugend (2023) *How can open innovation contribute to circular economy adoption? Insights from a literature review*	– How can open innovation contribute to the adoption of CE?	The authors encourage collaboration practices among partners and the adoption of co-creation approaches, to leverage open innovation through partnership networks between private and public institutions and local communities.	Meso level	Bibliometric and content analysis	2016–2021
Eisenreich et al. (2022) *Toward a circular value chain: Impact of the circular economy on a company's value chain processes*	– Which implications does the introduction of circular solutions have for a company's value chain processes?	Value chains need to be adjusted for addressing CE, considering how a) the linear production should be rethought to include multiple product life cycles, and b) the classic framework advanced by Porter does not sufficiently emphasise connections between value chain categories, nor acknowledge their interrelations with external stakeholders.	Firm level	Descriptive and content analysis	1990–2020
Sehnem et al. (2022) *Circular economy and innovation: a look from the perspective of organisational capabilities*	– What is the profile of publications dealing with CE and innovation? – How do researchers explore the existing interface between the concepts of CE and innovation? – What is the role of innovation in the transition to circular production models?	This study showcases how CE and innovation are related in the organisational context, outlining practices, organisational capabilities, types of innovation and management options adopted to implement CE in companies.	Firm level	Descriptive and content analysis	2016–2020
Sudusinghe and Seuring (2022) *Supply chain collaboration and sustainability performance in circular economy: a systematic literature review*	– What are the different collaboration practices leading to improved sustainability performance in circular supply chains? – What are the managerial and theoretical implications to ensure the implementation of circular supply chains through collaboration?	This study presents a framework for achieving sustainability performance through supply chain collaboration, highlighting how collaboration needs to evolve in supply chains. Furthermore, it outlines why the upstream and downstream perspectives on collaboration, as well as supply chain collaboration practices, need to be changed to address the CE transition.	Meso level	Descriptive and content analysis	2016–2020
Suchek et al. (2021) *Innovation and the circular economy: a systematic literature review*	The paper aims to provide insights into how innovation can drive the transition towards CE, outlining different research themes and emphasising the importance of cooperation and collaboration by highlighting internal and external factors to consider while addressing the CE transition.	Innovation in the CE depends on forming strategic alliances and taking a multi-level approach, incorporating all interested parties. These alliances enable the sharing of knowledge, raw materials, technology, and information crucial for companies to establish the conditions to capture the opportunities and develop innovation within the scope of the CE.	Firm and meso level	Bibliographic and content analysis	2016–2020

(Continues)

TABLE 1 | (Continued)

Article information	Research questions/Objectives	Outcomes and implications	Level of analysis	Review method	Range of years
Lahane et al. (2020) *Circular supply chain management: a state-of-art review and future opportunities*	The study highlights the research trends, gaps, and sets up future research directions in circular supply chain management literature.	The study recognises the importance of circular supply chain management in addressing the CE transition, and identifies potential areas for future research directions in this domain.	Firm and meso level	Descriptive and categorical analysis	2010–2019
Centobelli et al. (2020) *Designing business models in circular economy: a systematic literature review and research agenda*	The study aims to examine the current understanding related to designing business models within the context of a circular economy transition, in order to develop a research agenda that guides studies.	Through the identification of four main topic areas related to the design of circular business models, the study put emphasis on the role of policymakers and government agencies in promoting circular economy transitions, as well as the importance of leveraging technological, socio-cultural, economic, and institutional features to achieve successful circular economy transitions.	Firm level	Descriptive and content analysis	2013–2018
Farooque et al. (2019) *Industry 4.0 technologies as enablers of collaboration in circular supply chains: a systematic literature review*	The study aims to advance a new definition of circular supply chain management, map the related literature, and identify potential future avenues.	The article offers a comprehensive integrated view of circular supply chain management and deliver suggestions for future studies in terms of design for circularity, procurement and circular supply chain management, biodegradable packaging, supply chain collaboration and coordination, circular consumption, product liability and producers' responsibility, technologies adoption, drivers and barriers to circular supply chain development.	Firm level	Content analysis	2006–2018
De Angelis et al. (2018) *Supply chain management and the circular economy: towards the circular supply chain*	What are the implications for supply chain management in circular supply chains?	This study delivers insights about the potential elements related to CE that contribute to redefine supply chain management. It distinguishes traditional, linear, and circular supply chains; delivering a framework that yields insights into the CE and supply chain management.	Firm and meso level	Content analysis	2001–2017

Source: Authors' elaboration.

Finally, RQ3 is addressed by our comprehensive discussion that builds on the results from the systematic review to identify shortcomings and deliver future queries.

By clarifying different forms of inter-organisational collaboration and their attributes, our findings make three main contributions to the CE literature. First, we strengthen the understanding of collaborative initiatives underlying the CE transition by characterising circular strategic partnerships, circular supply chains, and circular ecosystem architectures (Brown et al. 2020; De Angelis et al. 2018; Kanda et al. 2021). Secondly, we recognised five thematic areas in terms of cross-cutting dimensions that demand further exploration in different collaborative patterns (Bocken and Ritala 2021; Brown, Von Daniels, et al. 2021; Chesbrough and Appleyard 2007; Enkel et al. 2009). Third, we developed several new research questions based on our insights to nourish the academic debate and open future research avenues.

As for practical implications, this study offers a valuable strategic compass to companies wishing to undertake a CE transition. Shedding light on peculiar inter-organisational collaboration patterns, we guide managers towards the strategic path that best suits each circular initiative. We also reiterate the importance of collaborating with homogeneous and heterogeneous actors to cope with resource shortages and overcome technical barriers. Finally, our study is instrumental in advancing the awareness of CE as a response to the Grand Challenges pointed out by policymakers and supra-governmental organisations.

2 | Methodology

A comprehensive and critical systematic literature review was conducted to answer the RQs. Systematic literature review is a methodology that allows existing studies to be rationalised and interpreted in order to offer an original point of view intended for stimulating new questions (Massaro et al. 2016; Kraus et al. 2020; Siddaway et al. 2019). We employed bibliometric and content analysis techniques to outline the main features of this literary segment and address thematic clustering (e.g., Bargoni et al. 2023; Bretas and Alon 2021; Vrontis and Christofi 2021). First, bibliographic coupling was employed to examine the literary roots of this literature niche and identify emergent research trends (e.g., Raut et al. 2023; Suchek et al. 2021). Then, we conducted a content analysis to systematise the available knowledge and shed light on each cluster to advance promising avenues for future research. In the intention of ensuring research transparency and replicability, this study complies with the notions delivered by the PRISMA statement (Moher et al. 2009; Page et al. 2021) and echoes the four-step protocol legitimised by previous systematic reviews (e.g., Hina et al. 2023; Hassan et al. 2023; Khan et al. 2021,b) that includes: planning the review, article screening criteria setting out, data extraction, data execution and research profiling.

2.1 | Planning the Review

The first step towards conducting a systematic literature review is the definition of the research protocol. To this end, it is imperative to assess what is known about a certain topic and identify a gap in the literature, acknowledging the need for a systematic review. Following the guidelines suggested by Kraus et al. (2020) and Massaro et al. (2016), a preliminary search on Google Scholar allowed us to gain an overview of the topic and evaluate previous systematic reviews about CE. We developed the search string accordingly, recognising the most frequently used keywords in the CE literature. By comparing these keywords with publications in leading academic journals on this topic and engaging in discussions with authors well-acquainted with the research topic, we further refined our keyword selection and assessed its exhaustivity. This process ensured the collection of as many studies as possible, closely related to inter-organisational collaborations and CE. Then, the final search string in Figure 1 was applied to the Scopus database, relying on previous studies that testify to Scopus' wider coverage than other scientific repositories in the social sciences' domain (Bargoni et al. 2023; Mongeon and Paul-Hus 2016; Singh et al. 2021).

2.2 | Screening Criteria

Once the search string and database for data collection were defined, the authors assessed the screening criteria of the literature review (e.g., Hina et al. 2023; Khan et al. 2021,b). The inclusion criteria required data to meet three requirements: (a) studies dwelling on inter-organisational collaboration of circular businesses; (b) peer-reviewed articles in press or published by highly ranked scientific journals (at least classified as Q2); (c) manuscripts written in the English language. On the other hand, exclusion criteria to discard the documents entailed: (a)

"innovat*" OR "collaborat*" OR "network*" OR "open*" OR "r&d" OR "system*" OR "ecosystem*"

AND

"circular economy" OR "ce" OR "bio-economy*" OR "bio economy*" OR "circular*"

AND

"business model*"

FIGURE 1 | Search string keywords.

research not strictly related to the CE domain or circular business models; (b) articles not pertaining to business management, social sciences, and environmental sciences subject areas; (c) articles and informative resources categorised as grey literature, thesis papers, editorials, book chapters and conference proceedings. In addition, since this study represents the first systematic review pursuing to systematise CE literature about inter-organisational collaboration, we decided to include every scientific article available from 2017 to 2023. Such a timeframe allowed us to include the most relevant studies and capture the growing scholarly interest. Overall, these screening criteria permitted the authors to follow a transparent and replicable analysis of the literature (Kraus et al. 2020; Massaro et al. 2016; Page et al. 2021; Tranfield et al. 2003).

2.3 | Data Extraction

The extraction process involved identifying, screening and selecting the articles for systematic review (see Figure 2). First thing, we executed the search string on the Scopus database covering "Article title, Abstract, Keywords", in January 2024. A total of 1.661 entries were identified. The results were thus filtered according to the outlined inclusion and exclusion criteria, obtaining 1.076 eligible papers for further screening.

With the intention of overcoming biases during data selection, two authors independently underwent the following two-step manual screening. They shared their findings at the end of the process to establish consensus. In the first step, they performed an early screening by examining the key elements of each record (i.e., article's title, keywords, and abstract). Documents irrelevant to the topic were discarded, resulting in the exclusion of most of the items. In the second step, the same authors went through the full text of the remaining articles to further sharpen the focus of the final selection. Wherever the two authors observed discrepancies in individual paper selection, they engaged in extensive discussions to resolve them and reach a consensus on the screening results. Eventually, this process led to the discarding of 974 records deemed inconsistent with the thematic boundaries of the study. As a result, 74 scientific articles that fall within the identified threshold of journal ranking were selected for data analysis.

Finally, we conducted a citation chaining search while reading the articles to identify any further documents not considered in

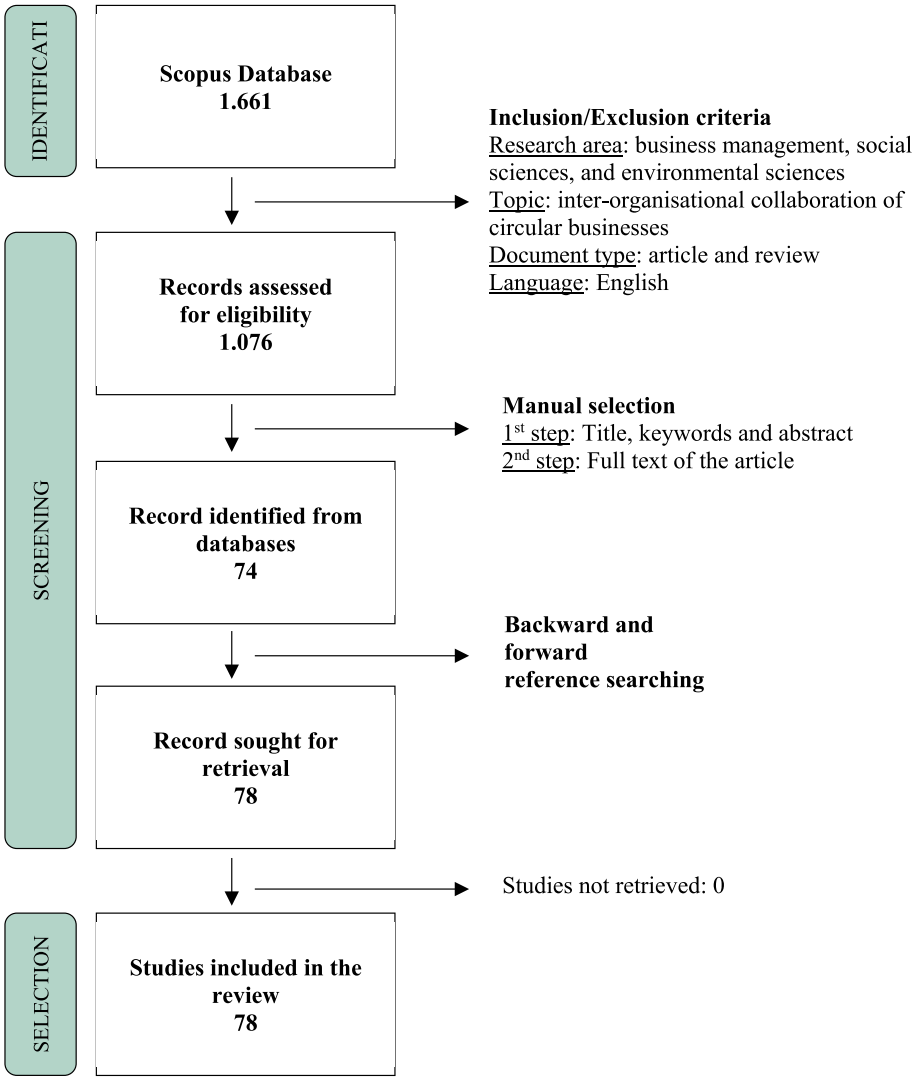


FIGURE 2 | Data extraction process.

the initial search (e.g. Chaudhary et al. 2023). Through forwards and backwards reference searching, upon debate and consensus among the authors, this step added 4 more articles to the final corpus for review. Hence, the final number of articles selected for the review rose to 78.

2.4 | Data Execution and Research Profiling

The selected papers were processed for the bibliometric and content analysis to explore the CE literature stream, recognise gaps in previous studies, and outline future avenues. Table 2 resumes the main information of the dataset, obtained using the Biblioshiny software tool (Aria and Cuccurullo 2017). From the final selection of 78 documents, we acknowledge a noticeable upward trend in the annual growth rate of scientific output (44.22%), a considerable number of authors involved in the CE literature stream about inter-organisational collaboration. Moreover, Figure 3 showcases that scientific production is concentrated in developed nations and some emerging countries.

TABLE 2 | Main information of the dataset.

Description	Result
Timespan	2017–2023
Sources	23
Documents	78
Annual growth rate	44.22%
Average citation per document	99
References	6.291
Authors	219

Source: Authors' elaboration using the Biblioshiny package in R.

Considering main authors' production over time (Figure 4), a greater concentration of work can be seen between 2019 and 2023, while the authors who have contributed most to the literature with impactful scientific articles are Nancy Bocken, Phil Brown, Ruud Balkenende, Gianmarco Bressanelli, Roberta De Angelis, Jan Konietzko, Leena Aarikka-Stenroos, Brian Baldassarre, and Anja Eisenreich. On the other hand, among the most frequent journals where the reviewed studies have been published, we can mention Business Strategy and the Environment; Journal of Cleaner Production; Resources, Conservation and Recycling; Sustainable Production and Consumption; International Journal of Production Economics; Journal of Business Research; International Journal of Production Research; and Journal of Industrial Ecology.

Observing the research profile of the sampled articles (see Appendix), most of the studies adopted a qualitative methodology to explore different collaboration strategies among companies and other entities through inductive reasoning. Previous authors mainly advanced multiple and single case study analysis, explorative research based on interviews and focus group interviews and data analysis based on the fuzzy Delphi methodology. This literature niche also includes systematic reviews on innovation, supply chain management, and collaboration enablers. Finally, a couple of papers addressed the topic from a deductive reasoning perspective to theoretical development employing structural equation modelling techniques. The reviewed documents also followed different lines in terms of theoretical positioning, including theoretical underpinnings such as resource-based view, dynamic capabilities, social capital theory, stakeholder theory, system thinking standpoint, organisational learning theory, organisational change theory, and organisational sensemaking theory.

Furthermore, we carried out a bibliographic coupling with the VOSviewer platform to assess the relationships between the reviewed scientific publications based on their shared

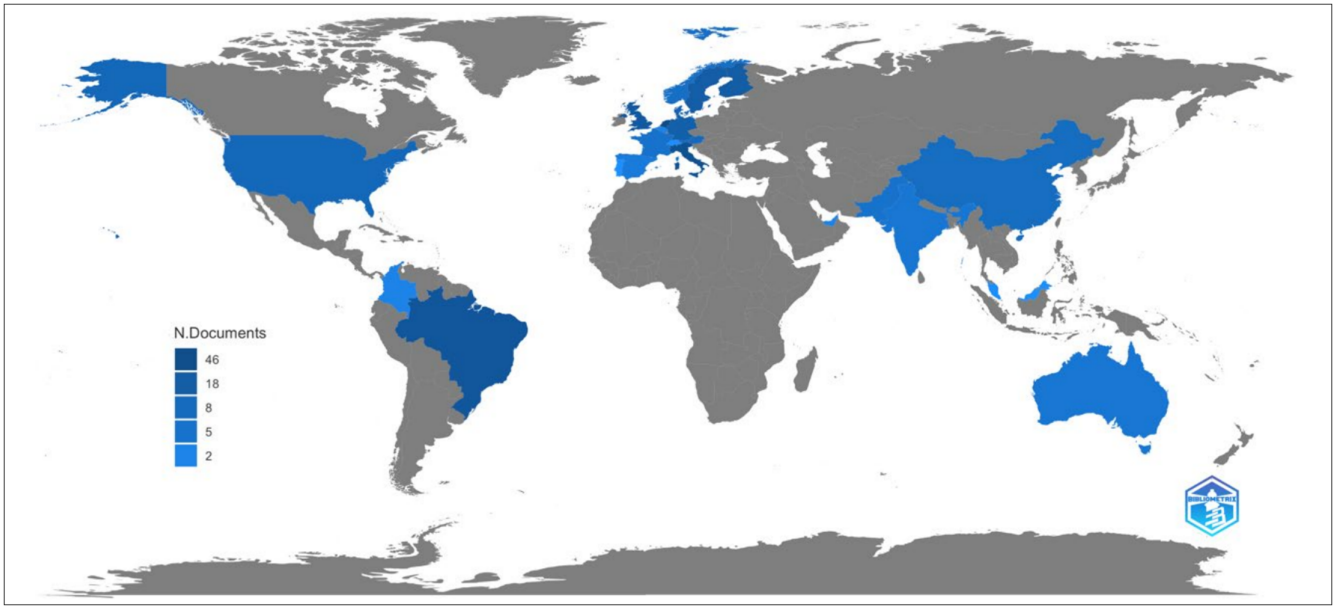


FIGURE 3 | Scientific production by country.

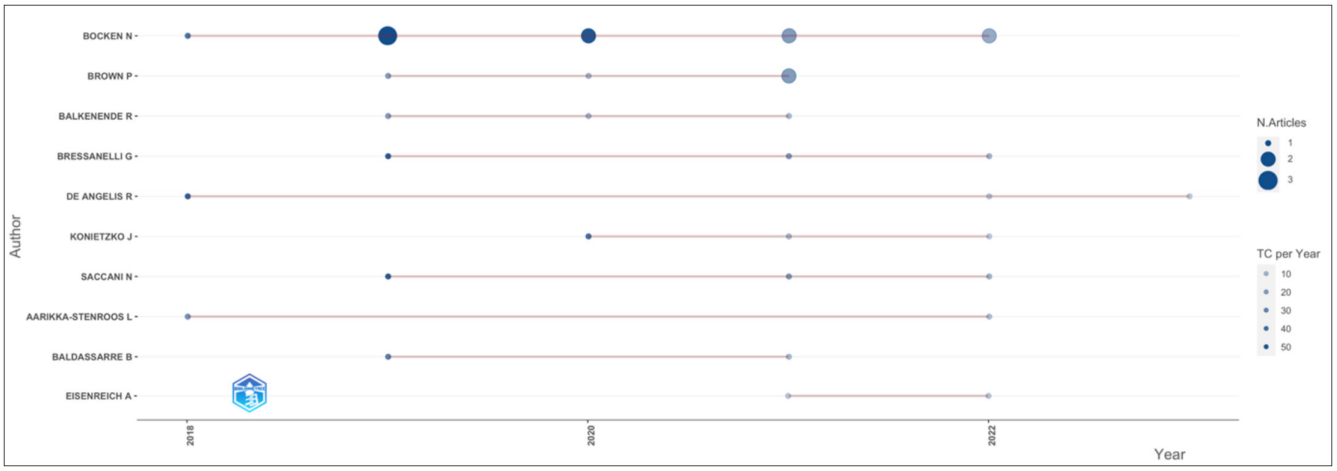


FIGURE 4 | Authors' production over time.

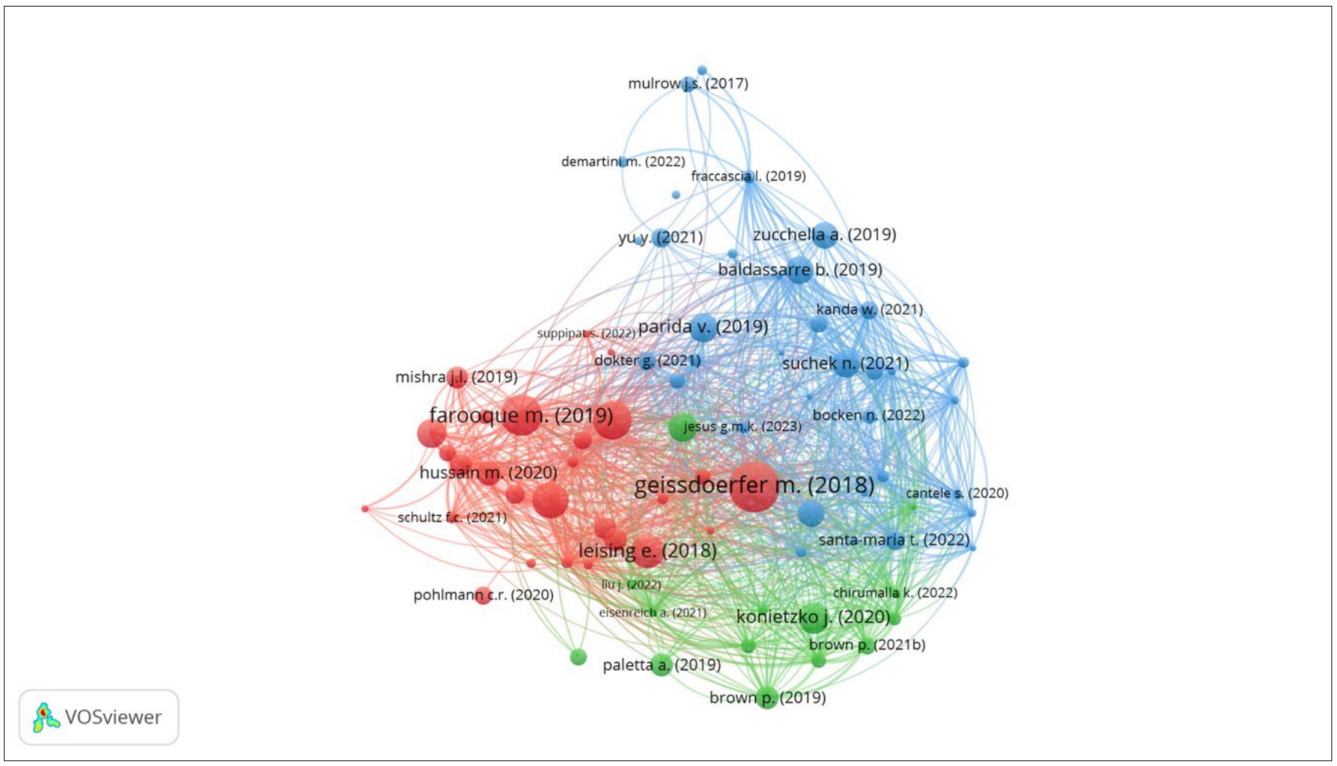


FIGURE 5 | Bibliographic coupling analysis.

references (Figure 5). According to previous studies (Boyack and Klavans 2010; Kessler 1963; Zupic and Čater 2015), we relied on this method to assess documents' similarity based on their shared references. With the intention of grouping items according to their literary derivations, we used this method to map a burgeoning research front by following the clustering strategy legitimised by past systematic reviews of the literature (e.g., Bartolacci et al. 2020; Raut et al. 2023). In the data analysis, papers with at least five citations in common with others were considered to ensure more reliable findings from analysing a database as broad as possible. As a result, we can outline three different clusters where 14 studies dwell on circular innovation management administered through strategic alliances (green), 27 contributions discuss supply chain management redefinition according to the CE dogmas (red), and 35 articles focus on ecosystem setting and industrial symbiosis establishment to address the CE transition (blue).

Based on the three literature roots identified with bibliographic coupling, the authors drew inspiration to advance the analysis of the thematic areas in this regard. More specifically, the two authors who underwent the previous manual screening process performed a thematic content analysis to outline recurring themes, identify the frequency and association between them, and interpret their significance (Braun and Clarke 2006; Neuendorf 2017). Thematic coding's objectivity and replicability were ensured by assessing intercoder reliability, in terms of agreement and consistency of the results obtained by the two

coders (O'Connor and Joffe 2020). Therefore, we undertook thematic content analysis by multiple coders to avoid potentially biased findings from a single interpretation. The next section offers the extensive findings of the content analysis, as a result of linking the output of the bibliometric analysis with the investigation of the recurring themes about inter-organisational collaboration.

3 | Findings From the Content Analysis

As a result of the thematic content analysis, we classify discernible collaborative approaches adopted by companies into three well-defined clusters. According to specific literary roots, we advance the conceptual categorization of circular strategic partnerships, circular supply chains, and circular ecosystems architectures. The remainder of this section is consequently intended to offer a thorough presentation of such collaborative approaches by outlining their literature foundations, main features, and recurring patterns in stakeholders' interaction while dealing with the CE transition.

3.1 | Circular Strategic Partnership

When it comes to project-specific partnerships thought to pursue CE principles implementation, it is essential to draw on literature dogmas on collaborative innovation management and the concept of sustainability, according to the triple bottom line (Brown et al. 2019; Elkington 1994; Suchek et al. 2021; Hidalgo and Albors 2008). On the one hand, collaborative innovation has been raised by companies as a response to complex problems or opportunities in the form of collective intelligence and shared resources to pursue more effective and holistic solutions than any single entity acting alone (Brown, Von Daniels, et al. 2021; Bryson et al. 2015; Wood and Gray 1991). This concept finds more concrete formalisation in open innovation strategies, emphasising the value of external knowledge sourcing, cross-industry partnerships based on resource exchange and value co-creation with external stakeholders (Chesbrough and Appleyard 2007; Enkel et al. 2009). On the other hand, circular project-specific collaborations are based on sustainable purposes. While pursuing R&D efforts that reconcile economic, social, and environmental goals, previous studies have labelled this specific kind of output as sustainable innovations (Boons and Lüdeke-Freund 2013; Cillo et al. 2019; Seebode et al. 2012). Then, specifically considering an approach where businesses and organisations actively collaborate with external stakeholders to co-create or develop innovations that promote sustainability, previous studies also referred to "sustainable open innovations" (Bogers et al. 2020, 1505; Kennedy et al. 2017). From this literature background, pursuing 'circular-oriented innovation' has been introduced as a specific approach to business sustainability that involves "the coordinated activities that integrate CE goals, principles, and recovery strategies into technical and market-based innovations" (Brown et al. 2019, 3). In other words, circular-oriented innovations involve an attempt to innovate products, processes, organisational practices and markets with the intent of slowing, narrowing and closing (material and energy) resource loops (Brown et al. 2020; Pieroni et al. 2019). As highlighted by Johnson (2022, 132), such a term can be used to describe "innovation processes for the circular initiatives", where "partner companies do not fall under a particular categorisation for a circular business model as they are still linear business models working with implementing circular initiatives". In this vein, circular-oriented innovations entail a strategic alliance that leverages open innovation to enable CE practices implementation, in the form of businesses' response to address the CE transition (Bocken and Ritala 2021; Eisenreich et al. 2021; Jesus and Jugend 2023; Köhler et al. 2022).

When pursuing the introduction of a circular innovation, organisations typically face tricky challenges in the form of complex and disruptive changes. As observed in multiple corporate sustainability reports by Stewart and Niero (2018, 1012), "several companies have initiated or engaged in active dialogue with, for example, peers, knowledge partners, value chain partners and regulators, to explore the role of CE in their specific business". Consequently, business initiatives typically engage in strategic partnerships to overcome the technical and structural barriers related to circular product or process development (Bressanelli et al. 2019; De Angelis et al. 2023; Suchek et al. 2021; Wrålsen et al. 2021). In agreement, Johnson (2022) observed that companies specifically seek stakeholders' collaboration to embrace circular oriented innovations due to a lack of capabilities and knowledge. Through such partnerships, firms gain access to tangible and intangible critical resources such as specific materials, machinery and space, knowledge or people skills, as well as competences related to governance and managerial structures to carry out circular oriented innovations (Boldrini and Antheaume 2021; Sehnem et al. 2022; Tapaninaho and Heikkinen 2022).

Another notable domain where stakeholders' collaboration is outlined as an 'inevitable element' is waste valorisation, leading to value creation in circular business models (Leder et al. 2020, 7). Specifically, we can recognise project-specific collaborations aimed at developing circular products or processes from previous empirical studies. For instance, Paletta et al. (2019) emphasised the importance of strategic partnership in the recycled plastic domain. The authors explain that manufacturing companies need to deal with stakeholders operating in waste management to dispose of recycled plastic; so plastic converters facilitate access to recovered resources according to customers' prerequisites. Another example is offered by Cantele et al. (2020), who reported the strategic partnership between a stationery supplier and an electrode producer to design and develop a pencil made from graphite powder waste. In this case, strategic collaboration proved instrumental in experimenting with the possibilities of remanufacturing processes. Circular product and process innovations provided a specific knowledge base about recovering and upcycling graphite, which was even employed in developing subsequent innovative one-to-one projects with fashion companies. Jensen et al. (2019) also investigated the implementation of remanufacturing practices undertaken in conjunction with a business partner, a sustainability-oriented firm experienced in remanufacturing processes. Overall, circular project-specific collaboration is a recurring inter-organisational pattern through which companies can carry out circular oriented innovations. According to this perspective, strategic partnerships typically involve business

entities aligned by the same vision to reduce the use of resources or energy in production cycles, as well as the waste resulting from their activities (Brown et al. 2019; Camilleri et al. 2023; Cantele et al. 2020; Tseng et al. 2022).

Dwelling on strategic alliances aimed at translating CE rationale into feasible practices, Brown et al. (2021,b) specifically outline six phases of the collaborative process establishment. In the first phase, a company identifies its needs in terms of knowledge, competences, tools and resources, to proceed later by communicating its intention to collaborate (Boldrini and Antheaume 2021; Johnson 2022). A careful selection of partners follows, with the aim of setting a shared vision on which to consequently structure collaborative processes and define the dynamics of value creation, delivery and capture (Pereira and Vence 2021; Ranta et al. 2018; Tapaninaho and Heikkinen 2022). Once these preliminary steps have been settled, businesses are required to undertake joint activities and collaborate on the implementation of circular oriented innovations (Brown et al. 2020; Eisenreich et al. 2021; Köhler et al. 2022; Stewart and Niero 2018). Jäger and Piscicelli (2021, 737) also showcased a 'collaboration set-up process', which includes five phases from motivation and need assessment, market and material flows understanding and vision alignment, the assessment of internal and external resources, partner evaluation and involvement, to the final agreement of every stakeholder.

Furthermore, strategic alliances based on specific innovation projects showed to be typically structured according to a defined time horizon toward the realisation of the circular product or process in question (Brown et al. 2019; Cantele et al. 2020; Pereira and Vence 2021). In this case, the partnership involves a limited number of companies, strictly related to the need for resources or capabilities required while carrying out the joint innovation project (Eisenreich et al. 2021; Jensen et al. 2019; Johnson 2022; Tapaninaho and Heikkinen 2022). The same companies involved are called upon to administer the governance of the collaboration, coordinating innovative efforts to develop circular products or processes. In contrast to the subsequent scenarios, in circular project-specific collaborations, it was possible to observe how there is not typically a predominant figure in charge of coordinating or orchestrating the strategic alliance (Brown, Von Daniels, et al. 2021; Cantele et al. 2020; Jensen et al. 2019; Paletta et al. 2019). The equal involvement of actors in decision-making and steering the strategic alliance is somewhat more pronounced, given the limited number of actors involved.

In conclusion, a circular strategic partnership represents an attempt to engage in the circular transition by a temporary alliance, involving a few companies around a specific innovation project. As suggested by the literature, this kind of strategic alliance lasts according to the duration of the circular project (Cantele et al. 2020; Jensen et al. 2019). Alternatively, it represents the spark that brings companies to include new entities in the creation of a larger scheme designed to generate new market opportunities by rethinking product value chains, or by establishing ecosystems that operate according to CE principles across a broad spectrum of activities (Carraresi and Bröring 2021; Kanda et al. 2021; Moggi and Dameri 2021; Schultz et al. 2021).

3.2 | Circular Supply Chain

The literary roots of circular supply chain management lie in various disciplines. Drawing on logistics and operations management literature (Bechtel and Jayaram 1997), previous authors introduced the concept of sustainable supply chain management and green supply chain management. Supply chain management, understood as information, product and capital flows optimisation within a system to maximise efficiency, responsiveness and resilience (Mentzer et al. 2001), has been further integrated by the triple bottom line's interpretation of sustainability (Elkington 1994). As such, sustainable supply chain management represents a holistic approach that aims to balance economic, social and environmental outcomes from supply chain activities (Jum'a et al. 2024; Seuring and Müller 2008). On the other hand, green supply chain management is a narrow approach to sustainable supply chain management that mainly addresses the environmental pillar of sustainability (Pohlmann et al. 2020; Srivastava 2007). It focuses on minimising the environmental impact of supply chain activities by resorting to environmentally friendly practices and strategies in the design, production, sourcing, logistics and distribution of products and services to achieve more environmentally sustainable operations. Accordingly, the literature outlined the circular supply chain as an additional configuration afferent to the broader concept of sustainable supply chain management (De Angelis et al. 2018; Farooque et al. 2019; Lahane et al. 2020).

The transition towards a CE brings traditional supply chains to be rethought according to sustainable opportunities, which can be specifically exploited by reducing resource depletion or recovering waste as new inputs in the value chain (Aarikka-Stenroos et al. 2022; Bressanelli et al. 2019; Eisenreich et al. 2022; Salmi and Kaipia 2022; Vegter et al. 2020). Then, it is possible to describe circular supply chains through the integration of CE principles in supply chain management, where the harmonisation of forward and reverse logistics allows multiple actors to establish a renewable system in a closed-loop value chain (Butt et al. 2023; Lahane et al. 2020; Tseng et al. 2022). Observing the coordinated adoption of CE practices by the actors gathered around the same value chain, the term circular supply chain has been introduced in literature as a collaborative strategic approach to support circular principles integration in businesses (Aray et al. 2023; Carraresi and Bröring 2021; De Angelis et al. 2018; Mishra et al. 2021; Tseng et al. 2022). Drawing on the definition advanced by Farooque et al. (2019), circular supply chain management can be described as "the integration of circular thinking into the management of the supply chain and its surrounding industrial and natural ecosystems". Thus, circular transition toward a renewable system in the supply chain dimension is based on industry specific inter-organisational collaboration (Ciulli et al. 2020; Gebhardt et al. 2022; Sudusinghe and Seuring 2022), where multiple independent actors commit to joint value chain activities designed for pursuing common objectives and mutual benefits (Cao and Zhang 2011; Soosay and Hyland 2015). Leising et al. (2018), investigating supply chain collaboration in the building sector, advocate that inter-organisational relationships are essential to create closed-loop supply chains that wink at a CE transition. In agreement, authors such as Bocken et al. (2016), Ghisellini et al. (2016), and Mishra et al. (2021) also emphasised scollaboration as a fundamental element in shaping

circular practices and processes. "When closing and slowing material loops, it is essential to include the supply chain as a whole, and to involve all parties from design and raw material suppliers to end users, service providers and recyclers, including the associated information flows" (Leising et al. 2018, 977). Questioning the role of designers in companies' circular transition, Dokter et al. (2021, 703) highlighted the 'multi-faceted challenge' represented by CE "that further expands the scope of design processes and projects". From the designers' side, the successful implementation of a circular business model demands cross-disciplinary knowledge arising from stakeholders' extensive collaboration in each value chain step. Indeed, strategic alliances that involve supply chain members are described in various studies as a critical requirement to succeed in the CE transition. For instance, Tseng et al. (2022, 743) outlined how "the collaboration within the value chain is seen as a critical facilitator for the transition process" in embracing a renewable production and consumption system. Drawing on Hussain and Malik (2020) and Rizos and Bryhn (2022), collaboration can be considered a process facilitator that enables circular transition in circular supply chain structures, besides supply chain configurations. The authors supported the link between inter-organisational collaboration and environmental performance of the supply chain, based on an "enhanced information and technology sharing in the supply chain network" (Hussain and Malik 2020, 9). As a result, Eisenreich et al. (2022, 9) advanced a 'circular value chain framework' as an adjustment of the value chain framework showcased by Porter (1985, 37), to better suit the cradle-to-cradle logic related to the CE domain and the interrelationships among the external stakeholders.

The literature echoes the concept of industrial symbiosis from a supply chain management perspective (e.g., Cavicchi and Vagnoni 2021; Mulrow et al. 2017), although a clarification is in order. As a collective approach based on geographically close companies pertaining to separate industries, industrial symbiosis involves the exchange of resources and materials to achieve a competitive advantage (Chertow 2007). This strategic approach, typically associated with system formations (Baldassarre et al. 2019; Fraccascia et al. 2019; Pavan et al. 2021), has been further expanded by Lombardi and Laybourn (2012, 31) according to a holistic view of intra-organisational collaboration, where companies aim to "foster eco-innovation and long-term culture change" by recurring to symbiosis. Coherently, it is possible to conceptualise industrial symbiosis in supply chain management as long as traditional and symbiotic actors from various sectors exchange resources and materials operating around a specific value chain (Turken and Geda 2020; Yu et al. 2021). In this vein, symbiotic actors concur to define a circular supply chain where companies pursue to close, narrow, and slow resource and energy loops (Bocken et al. 2016; De Angelis et al. 2018; Geissdoerfer et al. 2018).

In practice, companies can retrieve value from end-of-life products and recover scraps from transformation processes by engaging in a multi-stakeholder alliance in a circular supply chain. This inter-organisational formation has been recognised as a valuable strategy to recover value from waste, in order to establish a zero-waste supply chain by collaborating with industry's stakeholders (Carraresi and Bröring 2021; Cavicchi and Vagnoni 2021; Ciulli et al. 2020; Mishra et al. 2021). Coordinated procedures and mechanisms in a circular supply chain, or even involving multiple supply chains, allow residual waste or scraps from specific processes to become sources for another value chain step (Farooque et al. 2019; Leising et al. 2018; Tseng et al. 2022). Lahane et al. (2020) also supported that natural resources can be restored and regenerated to enhance their exploitation, thanks to accurate circular supply chain management. Therefore, strategic alliances are fundamental in circular supply chain management to establish circular practices and, building on the present review, we can advance some recurring patterns in these inter-organisational relationships. The collaboration mechanisms underlying circular supply chain functioning were categorised by Gebhardt et al. (2022) in information and resource sharing (Barratt 2004; Olorunniwo and Li 2010), mutual knowledge creation (Chan and Prakash 2012; Malhotra et al. 2005), joint planning and decision making (Barratt 2004; Olorunniwo and Li 2010), contractual and economic practices' formalisation (Cloutier et al. 2020).

Drawing on previous studies (e.g., Barratt 2004; Soosay and Hyland 2015; Sudusinghe and Seuring 2022), these practices can be arranged on vertical, horizontal, and lateral collaborations. They respectively represent interactions with upstream or downstream stakeholders, organisations at the same level of the supply chain or other external parties of competing supply chains, and the combination of the two forms. In addition, the transition toward a circular supply chain may require crossing industry boundaries, outlining an additional collaboration direction in the form of systemic interactions (Gebhardt et al. 2022). Schultz et al. (2021), agreeing with De Angelis et al. (2018) and Farooque et al. (2019), also support how circular supply chain development especially requires companies to emphasise the horizontal dimension to collaborate with multiple partners within and outside the core industry. Dealing with sustainable opportunities while pursuing the CE transition entails the interplay of more actors characterised by heterogeneous backgrounds (Aray et al. 2023; Tseng et al. 2022). Accordingly, the interaction of multiple stakeholders leads to revising traditional value chain formations, outdated for dealing with sustainable purposes. "Potential new specialised competitors, alone or in networks, enter the market with better practices, leading to new value chain emergence and in turn business model innovation, and triggering value redistribution among players" (Carraresi and Bröring 2021, 10). In this vein, circular businesses involve additional layers of complexity to supply chains, as well as the ownership of circular innovations and the distribution of costs and benefits among partners in the circular supply chain (Rizos and Bryhn 2022). Specifically referring to cross-industry partnerships, Carraresi and Bröring (2021) stress this point by emphasising how the establishment of new relationships with companies belonging to akin value chains improves companies' resilience in approaching a sustainable transition (Carayannis et al. 2014). In addition, another circularity facilitator has been found in supply chain configuration supported by structural flexibility of the actors in shaping material loops along the supply chain (De Angelis et al. 2018). Compared to a traditional value chain structure, the pronounced flexibility observed in circular supply chains supports businesses in addressing uncertainties and complexities related to supply chain management. In this vein, it allows sustainable product and process development, resource consumption minimisation and environmental technologies adoption (Bai et al. 2020; Liu et al. 2022).

Thus, the coordination of multiple actors along value chain stages allows to slow, narrow, close, intensify and dematerialise resource and energy loops (Chirumalla et al. 2022; De Angelis et al. 2018; Geissdoerfer et al. 2018; Leising et al. 2018). In general, Pohlmann et al. (2020) identified primary stakeholders as the actors directly involved in supply chain operations (e.g., manufacturers, suppliers, distributors, and consumers), while secondary stakeholders are typically embodied by organizations indirectly engaged in the supply chain (e.g., intermediaries, public entities, nongovernmental organizations, and media). Drawing on the case study advanced by Hansen and Revellio (2020), actors can find a specific categorisation in closed-loop supply chains according to their role. The authors highlighted manufacturers and retailers as central coordinators, typically positioned between suppliers and end users. Circular practices such as repair, reuse, remanufacture and recycle are carried out by the supply chain coordinator or loop operators, represented by support entities joining the circular supply chain. The coordinator role is covered by "forward-looking companies able to recognise opportunities in sustainability transition, to manage complementarities and to absorb disturbances going through necessary modifications to respond to disruptions" (Carraresi and Bröring 2021, 10). They act as focal companies within the circular supply chain, intended to coordinate the activities of other upstream or downstream actors, as well as those operators supporting loop strategies. Pohlmann et al. (2020) also pointed out the presence of focal companies that provide leadership and exercise the most significant control over decisions and activities in a circular supply chain. Their coordinating role involves designing the circular product or service offered, besides promoting triggers for supply chain management in terms of structuring organisations' involvement along the value chain (Pohlmann et al. 2020; Seuring and Müller 2008).

In summary, circular supply chain is framed in literature as an effective inter-organisational approach to address the CE transition. It entails the coordinated effort of multiple actors, within and beyond the main value chain, to establish closed-loop systems that promote economic, environmental, and social sustainability.

3.3 | Circular Ecosystem Architecture

The ecosystem perspective of circular businesses collaboration builds on different constructs from literature, including sustainable business models (Bocken et al. 2014), industrial ecology (Ashton 2008), industrial symbiosis (Chertow 2007), and business ecosystems (Iansiti and Levien 2004). A circular ecosystem architecture, at first, is established on the interaction of sustainable-oriented businesses, especially inclined to developing a closed and renewable system within the collaborative environment boundaries (Baldassarre et al. 2020; Bocken et al. 2014). On the other hand, industrial ecology offers some theoretical foundations for the CE discipline as a starting point for an ecosystem perspective. It advances the interpretation of "natural ecosystem as a metaphor for the design of industrial systems", where companies attempt to close, narrow, and slow resource and energy loops (Bocken et al. 2016; Kanda et al. 2021, 2815). In more detail, the industrial ecology branch of literature also explores industrial symbiosis as the establishment of a cooperative network of businesses based on resources and by-products exchange, as well as the common exploitation of infrastructure and services aimed at achieving competitive advantage besides environmental impacts reduction (Chertow 2007; Yu et al. 2021; Zaoual and Lecocq 2018). Drawing on the industrial ecology construct, actors' interaction in a renewable system supports the creation of a local ecosystem aggregation of organisations through an industrial symbiosis arrangement. As a strategic approach to establish inter-organisational collaboration, industrial symbiosis engages traditionally separate industries and actors in a systemic environment to capitalise on the synergistic possibilities offered by geographic proximity (Chertow 2007; Wolf et al. 2007). Baldassarre et al. (2019) employed industrial ecology in the CE domain as a conceptual lens to investigate a network of businesses based on industrial symbiosis, to outline actors' commonalities and differences. Further on, industrial symbiosis application in circular businesses has also been addressed by Fraccascia et al. (2019), as a system perspective to analyse the governance of a circular system characterised by numerous firms. In line with the business ecosystem properties, a CE ecosystem incorporates both production and consumption-side participants, including complementary asset providers and customers (Iansiti and Levien 2004; Yu et al. 2021). It allows circular ecosystems to stand out from alternative interorganisational formations, typically focused on either production or consumption-side participants (Kanda et al. 2021).

In general, we discussed how the implementation of CE practices triggers companies to cross their boundaries. Leaving a firm-centric focus, sustainability-oriented organisations are typically forced to interact with other actors in shaping a collaborative environment (Geissdoerfer et al. 2020; Kanda et al. 2021; Pieroni et al. 2019; Suppipat and Hu 2022). In this vein, an ecosystem architecture seeks to redefine companies' business models where multiple economic organisations, consumers, public entities, and institutions interact dynamically to create economic, social, and environmental value (Zaoual and Lecocq 2018; Zucchella and Previtali 2019). Konietzko et al. (2020) identified a circular ecosystem "as a set of actors—producers, suppliers, service providers, end users, regulators, civil society organizations—that contribute to a collective outcome" in achieving circularity. The ecosystem assessment highlights the interdependencies between different organisations and provides a lens to analyse value co-creation through systemic innovations. In this vein, Nogueira et al. (2023) outlined three cases where ecosystem innovation played a fundamental role in realising waste valorisation initiatives, proving that market innovation seems lacking in pursuing such sustainable opportunities. Tapaninaho and Heikkinen (2022) explored value creation in circular ecosystems, recognising stakeholders' evolving interactions as an effective mechanism to cope with the multiple facets of circular practices. Indeed, the authors found how "the local ecosystems are characterised by constantly evolving stakeholder relationships as new stakeholders join and others exit to seek optimal ecosystems and CE activities for their aims" (Tapaninaho and Heikkinen 2022, 2734). Also, Moggi and Dameri (2021) highlight how several stakeholders, which differ in industries and scope, can collaborate in a circular ecosystem to create shared value. The authors picture how actors' interactions support food waste reduction in an urban context by collecting, donating and redistributing

surplus food to a needy local community. More specifically, business ecosystems do not follow the linear value creation process and, thus, many of the actors in such collaborative environments are outside the scope of the traditional value creation chain setting (Barratt 2004; Iansiti and Levien 2004; Soosay and Hyland 2015). Heterogeneous entities and organisations, sometimes even competing with each other, take part in circular process or product development by gathering resources or delivering services to other actors. In this vein, members of a circular ecosystem deliver value through an interrelated system of interdependencies rather than as independent entities, horizontally broadening the boundaries of individual value chains to embrace realities and knowledge from different sectors (Nogueira et al. 2023; Tapaninaho and Heikkinen 2022).

Business ecosystem's orchestration capabilities, related to identifying and coordinating partners in a strategic network, were therefore found to be highly relevant in CE development (Parida et al. 2019; Geissdoerfer et al. 2020; Santa-Maria et al. 2022). Bringing together heterogeneous and previously unconnected actors from different industries, as well as other research entities, institutions or end consumers, stimulates out-of-the-box thinking to deal with sustainable opportunities from multiple angles (Konietzko et al. 2020).

According to natural ecosystems functioning, this form of collaboration is structured as a nested commercial system populated by character-specific players that concurs to an overarching solution or ecosystem-level goals coherent with CE principles (Christensen and Rosenbloom 1995; Nambisan and Baron 2013; Zaoual and Lecocq 2018). Zucchella and Previtali (2019) explored two issues related to the ecosystem's actors and their role in establishing circular processes and products. More specifically, the authors highlighted the presence of a focal actor, which behaves as an orchestrator of the circular network in sailing the other actors through the circular project. This figure typically generates trust and commitment among companies and entities involved in the collaborative network, while promoting communication about the shared vision and any opportunities in a win–win–win strategy (Elkington 1994; Suppipat and Hu 2022; Zaoual and Lecocq 2018). The authors introduced a second element that supports the development of circular practices, in terms of the ecosystem mechanisms related to actors' interactions (Zucchella and Previtali 2019). In this view, companies and institutions involved in a circular ecosystem cannot be identified as stand-alone entities; instead, they play a specific and fundamental role in network thriving. Drawing on Neutzling et al. (2018), the links among entities are based on tangible and intangible resource sharing. For instance, monetary investments and critical sharing of knowledge enable circular practices (Tapaninaho and Heikkinen 2022). Another critical aspect is actors' effective intention to collaborate, an element not to be underestimated considering organisations' heterogeneity as well as their specific needs and interests, sometimes even contrasting. Organisations are required to cooperate by looking beyond their individual interests to pursue the objectives of the whole system (Bressanelli et al. 2022; Parida et al. 2019). Finally, the third key element outlined by Zucchella and Previtali (2019) is represented by formal and informal governance agreements that regulate ecosystem mechanisms. Activities organisation and control, command structures and legal contracts are important conditions to regulate actors' activities in the network in the long term, as well as informal arrangements based on trust, communication, and cultural socialisation (Zucchella and Previtali 2019).

Consistent with a circular ecosystem architecture, integrating circular principles into business models requires a systemic view that encompasses the involvement of multiple stakeholders and their constructive interactions, (Moggi and Dameri 2021; Zucchella and Previtali 2019). Simultaneous collaboration and resource exchange among heterogeneous actors occurs, where new stakeholders join the ecosystem and others leave by designing constantly evolving intertwined relationships.

3.4 | Wrapping Up Circular Oriented Strategic Alliances

After a thorough systematisation of the literature, this section is intended to briefly summarise the differences between the three forms of inter-organisational collaboration recognised in previous studies. According to our findings, the authors reported in Table 3 the highlights resulting from the content analysis. Circular strategic partnerships, circular supply chain, and circular ecosystem architecture are categorised according to the scope and time horizon of the collaboration, partners involved, stakeholders' relationship structure, and governance.

4 | Discussion of the Results and Future Avenues

This systematic review represents the first attempt to classify and compare discernible collaborative approaches discussed in the CE literature. Although we recognised previous reviews' efforts in (a) shedding light on CE and innovation management by looking beyond firm boundaries (e.g., Jesus and Jugend 2023; Sehnem et al. 2022; Suchek et al. 2021), (b) delving into supply chain management and CE (Bressanelli et al. 2019; De Angelis et al. 2018; Farooque et al. 2019; Gebhardt et al. 2022; Hussain and Malik 2020; Lahane et al. 2020; Sudusinghe and Seuring 2022), (c) addressing CE alongside industrial symbiosis (Baldassarre et al., 2019; Demartini et al. 2022) or an ecosystem perspective (Puntillo et al. 2021); a fragmented and sometimes misleading awareness demanded systematisation. As a result, this study responds to the need for a clear compass guiding academics and practitioners towards a specific collaborative approach while dealing with the CE transition.

Drawing on our findings, we further stress the importance of inter-organisational collaboration in successfully addressing the CE transition. Strategic partnerships with stakeholders (e.g., Cooper and Claxton 2022; Jäger and Piscicelli 2021; Jensen et al. 2019; Paletta et al. 2019), alliances among companies from the same supply chain (e.g., Aarikka-Stenroos et al. 2022; Aray et al. 2023; Bressanelli et al. 2019; Carraresi and Bröring 2021; Ciulli et al. 2020; De Angelis et al. 2018; Mishra et al. 2021) or actors' collaboration in ecosystem architectures (e.g., Kanda et al. 2021; Moggi and Dameri 2021; Zaoual and Lecocq 2018; Zucchella and Previtali 2019) prove to foster business models' adaptation to a cradle-to-cradle economy. Inter-organisational collaboration, in different patterns, allows businesses to leverage

TABLE 3 | Wrapping circular economy-oriented collaborative approaches.

	Circular strategic collaboration	Circular supply chain	Circular ecosystem architecture
Scope	Focus on carrying out a specific circular oriented innovation	Focus on a specific value chain	Wide focus on multiple value chains and industries
Duration	Short-term perspective	Medium/long-term perspective	Long-term perspective
Stakeholders	Limited project-specific partners	Downstream and upstream stakeholders (mainly)	Cross-industry partnership
Structure	Semi-stable relationships among actors	Semi-stable relationships among actors	Evolving stakeholders' relationships network
Governance	Project-related partner agreement based on shared objectives and values	Focal actor as coordinator of the supply chain alliance Supply chain coordinators and loop operators	Focal actor as the orchestrator of the ecosystem
Stakeholders' affinity	Businesses suitable for project development	Businesses from the same product value chain, besides a few actors from the related ones	Actors from the same local context
Alliance ties	Temporary	Semi-persistent	Persistent
Involvement	Production side	Production side	Production and consumption side

Source: Authors' elaboration.

resources and critical knowledge sharing by addressing barriers toward the circular transition (Galvão et al. 2022; Perotti et al. 2024; Seles et al. 2022; Wade et al. 2022). As a result, collaborative initiatives such as circular strategic partnerships, circular supply chains, or circular ecosystem architectures represent a reliable approach to foster the transition toward a renewable production and consumption system.

Considering the reviewed articles (see Appendix), we found an uneven commitment to unravelling these collaborative approaches. Prior research has predominantly focused on circular supply chain management, with relatively limited attention given to the examination of circular ecosystems. Moreover, the most unexplored cluster concerns the unveiling of fixed-term strategic collaborations aimed at the development of specific projects. Seeing the current notable imbalance in the extent to which these three collaborative approaches have been explored in the existing body of literature, we encourage future studies to fill such a gap. Therefore, more studies are needed to properly comprehend these phenomena, specifically referring to circular ecosystems and project-specific collaborations aimed at developing circular-oriented innovations.

In addition, few studies have addressed multiple collaborative approaches simultaneously at the time of this systematic review (e.g., Perotti et al. 2024). Both inductive and deductive theorising are required to investigate and document similarities and differences of the outlined collaborative approaches. Academics and practitioners should acknowledge the implications of different collaborative settings in terms of stakeholders' engagement and resource exchange, as well as value creation and delivery from circular practices (e.g., Bressanelli et al. 2022; Civera et al. 2025; Moggi and Dameri 2021). Empirical validation is needed to characterise the different approaches and provide decision makers with the appropriate tools to choose the most suitable collaborative strategy. More specifically, the CE literature falls short in differentiating stakeholders' roles and interaction in circular strategic partnerships, circular supply chains, and circular ecosystems. Although previous studies highlighted the role of focal companies in circular networks (Konietzko et al. 2020; Perotti et al. 2024; Pohlmann et al. 2020), the figure of ecosystem orchestrator (Zaoual and Lecocq 2018; Zucchella and Previtali 2019) and supply chain coordinators (Carraresi and Bröring 2021; Hansen and Revellio 2020) should be better characterised, besides recognising new recurring figures in circular networks. Overall, stakeholders' contribution to the development of collaborative innovation aimed at tackling the CE transition has also been quite blurred so far. A focus on value creation, capture, and delivery in circular networks is also needed to further explore which kind of companies benefit more than others from these collaborations and to what extent (Cavicchi and Vagnoni 2021; de Vasconcelos Gomes et al. 2023; Kanda et al. 2021). In summary, a comparative assessment of shared value in collaborative networks is missing. Future research may assess the multidimensional value created and shared through the outlined collaborative networks (Tapaninaho and Heikkinen 2022).

The theme of collaboration is also closely linked to the exploitation of sustainable opportunities (Averina et al. 2022) through the development of circular-oriented innovations (Brown et al. 2020; Pereira and Vence 2021). More specifically, open innovation is echoed in each collaborative approach as a mechanism aimed at leveraging the sharing of knowledge, resources and instrumental assets to come up with circular product and processes (Calabrese et al. 2024; Eisenreich et al. 2021; Jesus and Jugend 2023; Perotti et al. 2024). In this respect, it would be

interesting to investigate how open innovation mechanisms are configured in the different approaches and their effectiveness. Previous authors such as Köhler et al. (2022) and De Angelis et al. (2023) referred to open innovation strategy in collaborative frameworks for CE by emphasising the role of dynamic capabilities. Drawing on Averina et al. (2022), in terms of competencies, partners can provide resources that are not available internally, or the company can contribute specific resources. It represents another interesting element that calls for further investigation to explore the role of individual and organisational capabilities in favouring collaborative innovation in circular networks (Santa-Maria et al. 2022; Sehnem et al. 2022).

Furthermore, some articles belonging to our sample (e.g., Blackburn et al. 2023; Ciulli et al. 2020; Langley et al. 2023; Leder et al. 2020; Liu et al. 2022) converge in elucidating how digital technologies can support circular transition by fostering interrelationships of actors within circular networks (Yu et al. 2022). Blackburn et al. (2023) introduced the role of digital platforms in supporting ecosystem's orchestration mechanisms, and Langley et al. (2023) investigated the role of artificial intelligence in optimising circular production within and among industrial ecosystems. Liu et al. (2022) explored digital technologies innovation besides supply chain management in the post-COVID economy, while Ciulli et al. (2020) assessed their supporting role in waste recovery considering a food supply chain. Overall, we recognise hidden value in exploring what technological innovation best suits a particular collaborative approach. Pursuing resource and information sharing, collaborative mechanisms coordination, value creation and distribution monitoring, academics and practitioners seek clarity and guidance in exploiting digital tools and artificial intelligence in circular strategic partnerships, circular supply chains and circular ecosystems.

Following our discussion, Table 4a and Table 4b resume the main gaps and potential research avenues inspired by our systematic literature review. Table 4a specifically showcases some main themes that future studies should untangle about circular

TABLE 4a | Literature gaps and potential research avenues: three collaboration patterns.

Theme	Gap	Future research avenues
Circular strategic partnership	Limited exploration of strategic collaborations aimed at developing circular oriented innovations to deal with specific projects	RQ1) What are the key success factors for effective project-specific collaborations in the context of circular economy? RQ2) How do project-specific collaborations differ from other collaborative approaches in terms of their impact on circular economy transition? RQ3) What mechanisms can facilitate knowledge sharing and resource allocation within project-specific collaborations for circular innovation?
Circular supply chain	Predominant focus on circular supply chain management with relatively limited attention to stakeholders' exchange of resources, and value creation and delivery in a closed-loop supply chain	RQ4) How can circular supply chains be optimised to enhance resource efficiency and minimise waste in the production and distribution process? RQ5) What are the barriers and enablers for the adoption of circular supply chain practices among different industries and sectors? RQ6) How do circular supply chains contribute to value creation and distribution among stakeholders in the circular economy? RQ7) What are the different roles assumed by the stakeholders in a circular supply chain? How can the duties of these figures be arranged best?
Circular ecosystem architecture	Limited exploration of circular ecosystems and stakeholders' role in supporting the circular economy transition	RQ8) How do circular ecosystem architectures facilitate collaboration and innovation among diverse stakeholders? RQ9) What are the characteristics of successful circular ecosystem architectures, and how can they be replicated in different local contexts? RQ10) What governance structures and mechanisms are effective in managing circular ecosystems for sustainable outcomes? RQ11) What are the different roles assumed by the stakeholders in a circular ecosystem architecture? How best to arrange the duties of these figures?

Source: Authors' elaboration.

TABLE 4b | Literature gaps and potential research avenues: cross-cutting dimensions.

Theme	Gap	Future research avenues
Value creation, delivery and capture	Incomplete understanding of how value is created and delivered within collaborative networks in the circular economy, as well as how to enhance value capture for the business	RQ12) What are the different dimensions of value created and shared through collaborative networks in the circular economy? RQ13) How do various collaborative approaches impact the distribution of costs and benefits among stakeholders in circular networks? RQ14) What mechanisms can ensure equitable and sustainable distribution of value among actors in different collaborative patterns?
Stakeholders' role and interaction	Limited characterisation of stakeholders' roles and interactions in different collaborative settings	RQ15) How do different stakeholders contribute to circular oriented innovation in different collaborative patterns, and what are their specific roles and responsibilities? RQ16) What are the motivations and incentives for stakeholders to participate in collaborative initiatives for the circular economy? RQ17) How can stakeholder engagement and participation be optimised to maximise the effectiveness of circular collaborations?
Open innovation	Limited exploration of how open innovation mechanisms are configured and implemented in different collaborative approaches	RQ18) What are the key drivers and barriers for implementing open innovation strategies in different circular networks? RQ19) How do open innovation mechanisms facilitate knowledge sharing and co-creation of value in different circular networks? RQ20) What are the implications of open innovation for intellectual property management and value capture in circular collaborations?
Dynamic capabilities	Limited understanding of the role of dynamic capabilities in facilitating collaborative innovation in different collaborative settings	RQ21) How do organisations develop and leverage dynamic capabilities to adapt to the changing requirements of circular collaborations? RQ22) What organisational processes and structures support the development and deployment of dynamic capabilities for circular innovation? RQ23) How can dynamic capabilities be measured and assessed in the context of circular collaborations to drive continuous improvement and innovation?
Digital technologies	Inadequate exploration of the role of digital technologies in supporting collaborative efforts and enabling circular practices	RQ24) How can digital technologies (such as blockchain, IoT, and AI) be leveraged to enhance collaboration and coordination within circular networks? RQ25) What are the challenges and opportunities associated with the adoption and integration of digital technologies in circular collaborations? RQ26) How do digital technologies contribute to data-driven decision-making and performance optimisation in different circular networks?

Source: Authors' elaboration.

strategic partnership, circular supply chain, and circular ecosystem architecture. Table 4b focuses on additional dimensions we encourage exploring across each collaborative pattern to acknowledge their implications in businesses. Both tables report a set of research questions, carefully crafted by drawing on the calls of the reviewed articles and our analysis.

Based on the elements highlighted in our discussion, we can also provide a conceptual model to support future quantitative studies. In Figure 6, we illustrate some of the factors influencing the implementation of CE in businesses, according to the business model dynamics of value creation, delivery and capture. As this systematic literature review identifies three inter-organisational collaborative approaches, we advance their moderating role in affecting the weight of each identified factor towards CE implementation. We expect the choice of one collaborative approach over another to directly influence open innovation mechanisms, emphasis on dynamic capabilities, the

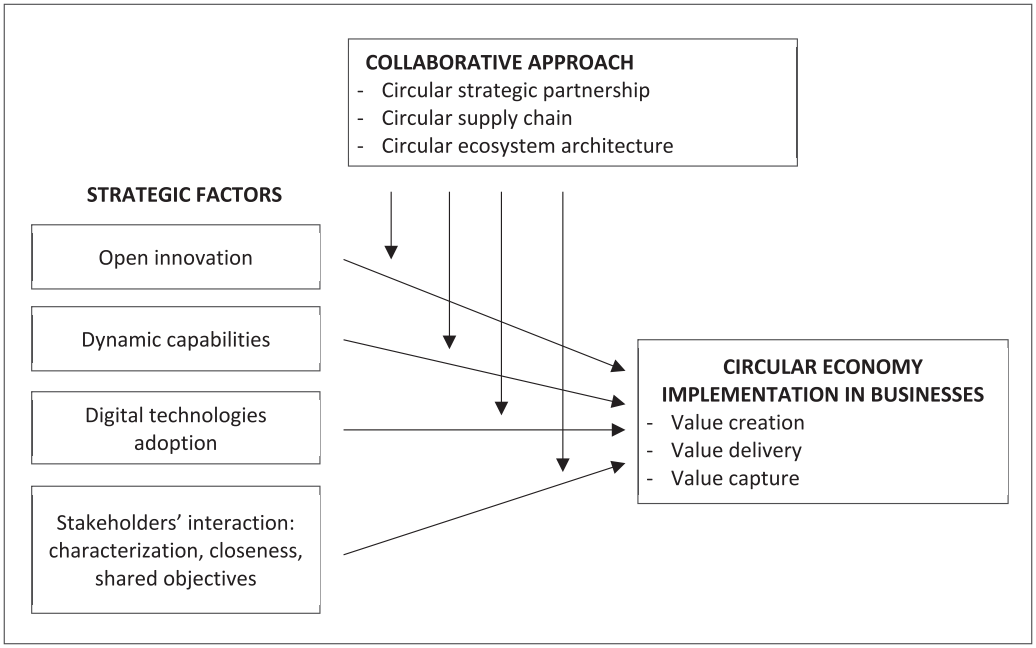


FIGURE 6 | Conceptual framework for future avenues.

instrumental use of digital technologies, or the entity and extent of stakeholder relationships.

5 | Contributions and Implications

This study represents an initial attempt to systematise the CE literature stream about inter-organisational collaboration, specifically focusing on a meso-level understanding of the phenomenon. Drawing on previous studies (see Appendix) that illuminate stakeholder collaboration's magnitude in navigating the complexities of the CE transition, our systematic review contributes to the literature debate as follows.

First, we shed light on distinguishable inter-organisational collaboration patterns by classifying circular strategic partnerships, circular supply chains and circular ecosystem architectures. Three primary literature roots were identified from bibliographic coupling, while the thematic content analysis' findings characterised each literature derivation in a specific collaborative pattern. Further enriching the perspective of previous studies (e.g., Carraresi and Bröring 2021; Parida et al. 2019; Perotti et al. 2024), we contribute to differentiating those collaborative approaches according to scope, stakeholders' nature and interaction mechanisms. By advancing a comprehensive understanding of inter-organisational collaboration in the CE domain, we also integrate previous studies that offered a narrowed perspective on collaborative innovation (e.g., Brown et al. 2020; Sehnem et al. 2022; Suchek et al. 2021; Jesus and Jugend 2023), circular supply chain alliances (e.g., De Angelis et al. 2018; Farooque et al. 2019; Lahane et al. 2020; Seuring and Müller 2008), industrial symbiosis (e.g., Baldassarre et al. 2019; Palagonia et al. 2025; Yu et al. 2021) or circular ecosystem (e.g., de Vasconcelos Gomes et al. 2023; Kanda et al. 2021; Konietzko et al. 2020; Moggi and Dameri 2021; Nogueira et al. 2023; Nambisan and Baron 2013). As a result, our study brings more clarity to the literature debate and opens new viewpoints to address CE by looking beyond organisational boundaries.

Second, we point out some key strategic dimensions that should be addressed along with circular strategic partnerships, circular supply chains, and circular ecosystem architectures. Arguing about five recurring themes retrieved from our thematic content analysis, we deliver a new perspective to complement previous studies on (a) value creation, delivery and capture dimensions (Cavicchi and Vagnoni 2021; Hansen and Revellio 2020; de Vasconcelos Gomes et al. 2023; Leder et al. 2020; Tapaninaho and Heikkinen 2022); (b) stakeholder role and interaction (Civera et al. 2025; Eisenreich et al. 2021; Moggi and Dameri 2021); (c) open innovation (Calabrese et al. 2024; Köhler et al. 2022; Perotti et al. 2024); (d) dynamic capabilities (De Angelis et al. 2023; Marrucci et al. 2022; Santa-Maria et al. 2022); and (e) digital technologies and AI (Blackburn et al. 2023; Ciulli et al. 2020; Langley et al. 2023). Based on these cross-cutting dimensions, we propose a conceptual framework to support academics, triggering further research and theoretical development in this field.

Third, we systematically present the major research gaps in this CE literature segment to pave the way for future research avenues. In this vein, after providing a comprehensive understanding of inter-organisational collaboration approaches and discussing some related thematic areas, we identify knowledge gaps pertinent to each theme. Out of the awareness gained throughout this systematic literature review, we practically contribute to the academic debate by proposing several research questions to inspire further studies.

Additionally, this research provides practical implications for businesses and policymakers. We emphasise inter-organisational collaboration as a viable approach to deal with the CE transition, besides clarifying different collaborative approaches for practitioners. Chief sustainability officers, sustainability specialists

and other managerial figures may use our insights to choose the most suitable collaborative approach to develop circular products, processes or practices. Specifically referring to the focal actors of circular supply chains or circular ecosystems, our findings suggest that managers should design supportive policies and incentives that encourage and facilitate integration among firms, non-profit entities and public actors. Furthermore, we provide managers with suggestions about strategic dimensions to consider for successful CE implementation.

Finally, policymakers like the European Commission can draw on our results to implement sustainable development policies that promote inter-organisational collaboration among actors. An example is represented by the European Partnerships, an instrumental approach promoted by the European Research and Innovation Framework Programme, Horizon Europe, and the European Research Area, which bring together private and public entities to deal with Europe's most pressing challenges (European Commission 2025). Such programs can be specifically adapted to promote the CE transition by leveraging inter-organisational collaboration.

6 | Conclusive Remarks and Limitations

In conclusion, our systematic review of the literature provides valuable insights into the current state of the art of inter-organisational collaboration in the CE domain. By highlighting existing knowledge gaps and providing implications for both research and practice, we support academics, businesses and decision-makers in their attempt to navigate the transition towards a more sustainable and circular economic system.

As a preliminary remark, our findings point out the early stage of the CE literature branch studying inter-organisational collaboration, mainly exploring phenomena through the abstraction of empirically grounded knowledge and offering content systematisation. Future studies may adopt deductive reasoning to theory building by assessing previous findings through empirical observation on representative samples. In line with previous authors' recommendations (e.g., Averina et al. 2022; Butt et al. 2023; Ciulli et al. 2020; Lahane et al. 2020; Liu et al. 2022; Pieroni et al. 2019; Santa-Maria et al. 2022; Sehnem et al. 2022), we emphasise that future studies should focus on quantitative approaches to empirically assess potential relationships among key variables involved in CE transition, and enhance the generalisability of findings across industries and geographical contexts.

Then, the main limitations of this study are mostly related to the choices made when designing the methodology. As an example, the decision to rely on a single database allowed us to conduct bibliometric analyses with consistent data on scientific publications, although this may represent the loss of some valuable contributions from other scientific sources. Considering how the tendency to extract peer-reviewed articles from scientific databases is based on quality and validity concerns, we may also have missed interesting items among other document types such as book chapters, conference proceedings, non-peer-reviewed papers, and practice-based journals. Furthermore, even though we resorted to forward and backwards reference searching, this method suffers from poor randomised representativeness. Despite our efforts, we might have overlooked some items due to the different indexing of the databases. Finally, while the lack of empirical validation can represent a limitation at this stage, this systematic literature review delivers several valuable suggestions for future studies.

Author Contributions

Francesco Antonio Perotti: conceptualisation; methodology; data curation; formal analysis; visualisation; writing – original draft preparation; writing – review and editing. **Ilan Alon:** methodology; resources; supervision; review and editing (critical review and commentary). **Bhumika Gupta:** review and editing (critical review and commentary). **Alberto Ferraris:** conceptualization; methodology; supervision; validation; review and editing (critical review and commentary).

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Appendix A

Authors and publication year	Title	Journal (Quartile)	Theoretical underpinnings	Methodology
Aray et al. (2023)	*Integrating closed-loop principles in supply chains in emerging markets: The case of the Russian waste management industry*	European Management Review (Q1)	Institutional Theory	Multiple case study analysis
Butt et al. (2023)	*The role of reverse logistics in a circular economy for achieving sustainable development goals: a multiple case study of retail firms*	Production Planning and Control (Q1)	—	Multiple case study analysis
Nogueira et al. (2023)	*From waste to market: Exploring markets, institutions, and innovation ecosystems for waste valorisation*	Business Strategy and the Environment (Q1)	—	Multiple case study analysis
De Angelis et al. (2023)	*Open strategy and dynamic capabilities: A framework for circular economy business models research*	Business Strategy and the Environment (Q1)	—	Conceptual study: explanatory theorising
de Vasconcelos et al. (2023)	*Circular ecosystem management: Orchestrating ecosystem value proposition and configuration*	International Journal of Production Economics (Q1)	—	Multiple case study analysis
Langley et al. (2023)	*Orchestrating a smart circular economy: Guiding principles for digital product passports*	Journal of Business Research (Q1)	—	Conceptual study: trans-disciplinary analysis
Jesus and Jugend (2023)	*How can open innovation contribute to circular economy adoption? Insights from a literature review*	European Journal of Innovation Management (Q1)	—	Systematic literature review: bibliometric and content analyses
Cavicchi and Vagnoni (2021)	*The role of performance measurement in assessing the contribution of circular economy to the sustainability of a wine value chain*	British Food Journal (Q1)	—	Single case study analysis
Averina et al. (2022)	*Assessing sustainability opportunities for circular business models*	Business Strategy and the Environment (Q1)	—	Multiple case study analysis
Gebhardt et al. (2022)	*Industry 4.0 technologies as enablers of collaboration in circular supply chains: a systematic literature review*	International Journal of Production Research (Q1)	—	Systematic literature review: content analysis
De Angelis (2022)	*Circular economy business models as resilient complex adaptive systems*	Business Strategy and the Environment (Q1)	Systems theory	Integrative research approach
Köhler et al. (2022)	*Towards a collaboration framework for circular economy: The role of dynamic capabilities and open innovation*	Business Strategy and the Environment (Q1)	Dynamic capabilities	Single case study analysis
Pedersen et al. (2023)	*Navigating value networks to co-create sustainable business models: An actionable staging approach*	Business Strategy and the Environment (Q1)	Collaborative design and actor-network theory	Single case study analysis
Santa-Maria et al. (2022)	*How do incumbent firms innovate their business models for the circular economy? Identifying micro-foundations of dynamic capabilities*	Business Strategy and the Environment (Q1)	Dynamic capabilities	Multiple case study analysis
Sehnem et al. (2022)	*Circular economy and innovation: A look from the perspective of organizational capabilities*	Business Strategy and the Environment (Q1)	Theory of innovation	Systematic literature review: bibliometric and content analyses

Authors and publication year	Title	Journal (Quartile)	Theoretical underpinnings	Methodology
Tapaninaho and Heikkinen (2022)	*Value creation in circular economy business for sustainability: A stakeholder relationship perspective*	Business Strategy and the Environment (Q1)	Stakeholder theory	Single case study analysis
Demartini et al. (2022)	*An investigation into modelling approaches for industrial symbiosis: A literature review and research agenda*	Cleaner Logistics and Supply Chain (Q1)	—	Systematic literature review: bibliometric and content analyses
Aarikka-Stenroos et al. (2022)	*Companies' circular business models enabled by supply chain collaborations: An empirical-based framework, synthesis, and research agenda*	Industrial Marketing Management (Q1)	—	Multiple case study analysis
Tseng et al. (2022)	*Healthcare industry circular supply chain collaboration in Vietnam: vision and learning influences on connection in a circular supply chain and circularity business model*	International Journal of Logistics Research and Applications (Q1)	Organisational learning theory, resource-based view, organisational change	Fuzzy Delphi method and decision-making trial and evaluation laboratory
Bressanelli et al. (2022)	*Circular Economy and the evolution of industrial districts: a supply chain perspective*	International Journal of Production Economics (Q1)	—	Multiple case study analysis
Sudusinghe and Seuring (2022)	*Supply chain collaboration and sustainability performance in circular economy: A systematic literature review*	International Journal of Production Economics (Q1)	—	Systematic literature review: content and contingency analyses
Chirumalla et al. (2022)	*Mapping a circular business opportunity in electric vehicle battery value chain: A multi-stakeholder framework to create a win-win-win situation*	Journal of Business Research (Q1)	—	Research-through-design approach
Bocken and Ritala (2021)	*Six ways to build circular business models*	Journal of Business Strategy (Q1)	—	Conceptual study
Cooper and Claxton (2022)	*Garment failure causes and solutions: Slowing the cycles for circular fashion*	Journal of Cleaner Production (Q1)	—	Visual analysis based on interviews, focus groups, round tables, and an expert workshop
Eisenreich et al. (2022)	*Toward a circular value chain: Impact of the circular economy on a company's value chain processes*	Journal of Cleaner Production (Q1)	—	Systematic literature review: bibliometric and content analyses
Rizos and Bryhn (2022)	*Implementation of circular economy approaches in the electrical and electronic equipment (EEE) sector: Barriers, enablers and policy insights*	Journal of Cleaner Production (Q1)	—	Multiple case study analysis
Salmi and Kaipia (2022)	*Implementing circular business models in the textile and clothing industry*	Journal of Cleaner Production (Q1)	Dynamic capabilities	Multiple case study analysis
Liu et al. (2022)	*Digital technologies and circular economy in supply chain management: in the era of COVID-19 pandemic*	Operations Management Research (Q1)	Dynamic capabilities	Partial least squares structural equation modelling
Blackburn et al. (2023)	*Digital Platforms for the Circular Economy: Exploring Meta-Organizational Orchestration Mechanisms*	Organization and Environment (Q1)	—	Multiple case study analysis
Marrucci et al. (2022)	*Using a life cycle assessment to identify the risk of "circular washing" in the leather industry*	Resources, Conservation and Recycling (Q1)	—	Single case study analysis and life cycle assessment

Authors and publication year	Title	Journal (Quartile)	Theoretical underpinnings	Methodology
Suppipat and Hu (2022)	*Achieving sustainable industrial ecosystems by design: a study of the ICT and electronics industry in Taiwan*	Journal of Cleaner Production (Q1)	—	Qualitative study based on interviews
Han et al. (2022)	*How do companies launch circular service business models in different countries?*	Sustainable Production and Consumption (Q1)	—	Qualitative study based on interviews
Johnson (2022)	*Closing competency gaps for circularity: Exploring partner dynamics for circular-oriented innovation*	Sustainable Production and Consumption (Q1)	Resource-based view, organisational learning	Single case study analysis
Kanda et al. (2021)	*From circular business models to circular business ecosystems*	Business Strategy and the Environment (Q1)	—	Multiple case study analysis
Moggi and Dameri (2021)	*Circular business model evolution: Stakeholder matters for a self-sufficient ecosystem*	Business Strategy and the Environment (Q1)	Stakeholder theory	Single case study analysis
Puntillo et al. (2021)	*Reevaluating waste as a resource under a circular economy approach from a system perspective: Findings from a case study*	Business Strategy and the Environment (Q1)	—	Systematic literature review and single case study analysis
Suchek et al. (2021)	*Innovation and the circular economy: A systematic literature review*	Business Strategy and the Environment (Q1)	—	Systematic literature review: bibliographic and content analyses
Boldrini and Antheaume (2021)	*Designing and testing a new sustainable business model tool for multi-actor, multi-level, circular, and collaborative contexts*	Journal of Cleaner Production (Q1)	—	Single case study analysis based on experiments and workshops
Bressanelli et al. (2021)	*Enablers, levers and benefits of Circular Economy in the Electrical and Electronic Equipment supply chain: a literature review*	Journal of Cleaner Production (Q1)	—	Systematic literature review: bibliographic and content analyses
Brown, Baldassarre, et al. (2021)	*A tool for collaborative circular proposition design*	Journal of Cleaner Production (Q1)	—	Design science research
Brown, Von Daniels, et al. (2021)	*A process model for collaboration in circular oriented innovation*	Journal of Cleaner Production (Q1)	—	Single case study analysis
Carraresi and Bröring (2021)	*How does business model redesign foster resilience in emerging circular value chains?*	Journal of Cleaner Production (Q1)	—	Single case study analysis
Pavan et al. (2021)	*Circular business models for bioelectricity: A value perspective for sugar-energy sector in Brazil*	Journal of Cleaner Production (Q1)	—	Single case study analysis
Pereira and Vence (2021)	*The role of KIBS and consultancy in the emergence of Circular Oriented Innovation*	Journal of Cleaner Production (Q1)	—	Qualitative study based on interviews
Schultz et al. (2021)	*Circular supply chain governance: A qualitative-empirical study of the European polyurethane industry to facilitate functional circular supply chain management*	Journal of Cleaner Production (Q1)	—	Qualitative study based on interviews
Wrålsen et al. (2021)	*Circular business models for lithium-ion batteries-Stakeholders, barriers, and drivers*	Journal of Cleaner Production (Q1)	—	Delphi method
Yu et al. (2021)	*Towards Circular Economy through Industrial Symbiosis in the Dutch construction industry: A case of recycled concrete aggregates*	Journal of Cleaner Production (Q1)	Complex adaptive system theory	Single case study analysis based on agent-based modelling

Authors and publication year	Title	Journal (Quartile)	Theoretical underpinnings	Methodology
Eisenreich et al. (2021)	*Open Circular Innovation: How Companies Can Develop Circular Innovations in Collaboration with Stakeholders*	Sustainability (Q2)	Stakeholder theory	Qualitative study based on interviews
Dokter et al. (2021)	*How circular is current design practice? Investigating perspectives across industrial design and architecture in the transition towards a circular economy*	Sustainable Production and Consumption (Q1)	—	Qualitative study based on interviews
Jäger and Piscicelli (2021)	*Collaborations for circular food packaging: The set-up and partner selection process*	Sustainable Production and Consumption (Q1)	—	Delphi method
Ciulli et al. (2020)	*Circularity Brokers: Digital Platform Organizations and Waste Recovery in Food Supply Chains*	Journal of Business Ethics (Q1)	Social network theory	Qualitative study based on interviews
Hussain and Malik (2020)	*Organizational enablers for circular economy in the context of sustainable supply chain management*	Journal of Cleaner Production (Q1)	Organisational sensemaking theory	Systematic literature review and structural equation modelling
Konietzko et al. (2020)	*Circular ecosystem innovation: An initial set of principles*	Journal of Cleaner Production (Q1)	—	Pattern matching and single case study analysis
Lahane et al. (2020)	*Circular supply chain management: A state-of-art review and future opportunities*	Journal of Cleaner Production (Q1)	—	Systematic literature review: content analysis
Pohlmann et al. (2020)	*The role of the focal company in sustainable development goals: A Brazilian food poultry supply chain case study*	Journal of Cleaner Production (Q1)	—	Systematic literature review and single case study analysis
Hansen and Revellio (2020)	*Circular value creation architectures: Make, ally, buy, or laissez-faire*	Journal of Industrial Ecology (Q1)	Transaction cost economics, resource-based view	Multiple case study analysis
Leder et al. (2020)	*Influential factors for value creation within the Circular Economy: Framework for Waste Valorisation*	Resources, Conservation and Recycling (Q1)	Social capital theory	Systematic literature review and qualitative study based on focus groups
Vegter et al. (2020)	*Supply chains in circular business models: processes and performance objectives*	Resources, Conservation and Recycling (Q1)	—	Systematic literature review: content analysis
Brown et al. (2020)	*How do companies collaborate for circular oriented innovation?*	Sustainability (Q2)	—	Qualitative study based on interviews
Cantele et al. (2020)	*Spreading sustainability innovation through the co-evolution of sustainable business models and partnerships*	Sustainability (Q2)	—	Single case study analysis
Zucchella and Previtali (2019)	*Circular business models for sustainable development: A "waste is food" restorative ecosystem*	Business Strategy and the Environment (Q1)	—	Single case study analysis
Bressanelli et al. (2019)	*Challenges in supply chain redesign for the Circular Economy: a literature review and a multiple case study*	International Journal of Production Research (Q1)	—	Systematic literature review and multiple case study analysis
Parida et al. (2019)	*Orchestrating industrial ecosystem in circular economy: A two-stage transformation model for large manufacturing companies*	Journal of Business Strategy (Q2)	—	Multiple case study analysis

Authors and publication year	Title	Journal (Quartile)	Theoretical underpinnings	Methodology
Baldassarre et al. (2019)	*Industrial Symbiosis: towards a design process for eco-industrial clusters by integrating Circular Economy and Industrial Ecology perspectives*	Journal of Cleaner Production (Q1)	—	Single case study analysis
Farooque et al. (2019)	*Circular supply chain management: A definition and structured literature review*	Journal of Cleaner Production (Q1)	—	Systematic literature review: bibliometric and content analyses
Jensen et al. (2019)	*Creating sustainable value through remanufacturing: Three industry cases*	Journal of Cleaner Production (Q1)	—	Multiple case study analysis
Mishra et al. (2021)	*Collaboration as an enabler for circular economy: a case study of a developing country*	Management Decision (Q1)	Natural resource-based view	Single case study analysis
Paletta et al. (2019)	*Barriers and challenges to plastics valorisation in the context of a circular economy: Case studies from Italy*	Journal of Cleaner Production (Q1)	—	Quantitative and qualitative analyses
Fraccascia et al. (2019)	*Business models for industrial symbiosis: A taxonomy focused on the form of governance*	Resources, Conservation and Recycling (Q1)	—	Conceptual study
Brown et al. (2019)	*Why do companies pursue collaborative circular oriented innovation?. Sustainability, 11(3), 635.*	Sustainability (Q2)	—	Multiple case study analysis
Stewart and Niero (2018)	*Circular economy in corporate sustainability strategies: A review of corporate sustainability reports in the fast-moving consumer goods sector*	Business Strategy and the Environment (Q1)	—	Systematic review of corporate reports: content analysis and mapping approach
Zaoual and Lecocq (2018)	*Orchestrating Circularity within Industrial Ecosystems: Lessons from iconic cases in three different countries*	California Management Review (Q1)	—	Multiple case study analysis
Geissdoerfer et al. (2018)	*Business models and supply chains for the circular economy*	Journal of Cleaner Production (Q1)	—	Multiple case study analysis
Leising et al. (2018)	*Circular Economy in the building sector: Three cases and a collaboration tool*	Journal of Cleaner Production (Q1)	—	Multiple case study analysis
Neutzling et al. (2018)	*Linking sustainability-oriented innovation to supply chain relationship integration*	Journal of Cleaner Production (Q1)	—	Multiple case study analysis
Ranta et al. (2018)	*Creating value in the circular economy: A structured multiple-case analysis of business models*	Journal of Cleaner Production (Q1)	—	Multiple case study analysis
De Angelis et al. (2018)	*Supply chain management and the circular economy: towards the circular supply chain*	Production Planning and Control (Q1)	—	Systematic literature review: content analysis
Mulrow et al. (2017)	*Industrial Symbiosis at the Facility Scale*	Journal of Industrial Ecology (Q1)	—	Conceptual study